

A Comparison between Multiple Linear Regression and Autoregressive Models

In multiple linear regression we have the response variable y and the m independent variables $x_1, x_2, \dots, x_m$. For each of the n observations, we have

$$y_i = a_0 + a_1x_{i1} + a_2x_{i2} + \dots + a_mx_{im} + z_i, \quad (1)$$

where z_i is the error term. We normally assume that the errors are independently and identically distribute Normal $N(0, \sigma)$ but in the analysis below, it can be done without actually assuming the distribution of the error terms. Writing (1) in vector matrix format, we obtain

$$\begin{aligned} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} &= a_0 \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + a_1 \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix} + a_2 \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} + \dots + a_m \begin{bmatrix} x_{1m} \\ x_{2m} \\ \vdots \\ x_{nm} \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \\ &= \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \dots & \dots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}. \end{aligned} \quad (2)$$

From (2), we can rewrite it as

$$y = Xa + z, \quad (3)$$

$$\text{where } y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \dots & \dots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{bmatrix}, a = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \text{ and } z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}.$$

We assume that none of the columns in X are linearly depending on other columns.

Some preliminaries from Matrix and Linear Algebra which we need to know. Firstly note that for any square matrix M , the expression $x^\top Mx$ is 1-dimensional and is known as a quadratic form. A matrix M is positive definite if $x^\top Mx \geq 0$ for any x . The matrix $X^\top X$ is square and is positive definite (if none of the columns are linearly dependent on other columns), and so is $(X^\top X)^{-1}$.

From (3), we now have $z = y - Xa$ and the sum of squared errors is

$$\begin{aligned}
z^\top z &= (y - Xa)^\top (y - Xa) \\
&= (y^\top - a^\top X^\top)(y - Xa) \\
&= y^\top y - a^\top X^\top y - y^\top Xa + a^\top (X^\top X)a \\
&= a^\top (X^\top X)a - a^\top X^\top y - a^\top X^\top y + y^\top y \\
&= a^\top (X^\top X)a - 2a^\top X^\top y + y^\top y.
\end{aligned} \tag{4}$$

Note that $a^\top X^\top y = (a^\top X^\top y)^\top = y^\top Xa$ since it is 1×1 , that is, 1-dimensional.

Now we add and subtract $y^\top X(X^\top X)^{-1}X^\top y$ to the right hand side of (4) to obtain

$$\begin{aligned}
z^\top z &= a^\top (X^\top X)a - 2a^\top X^\top y + y^\top X(X^\top X)^{-1}X^\top y + y^\top y - y^\top X(X^\top X)^{-1}X^\top y \\
&= a^\top (X^\top X)a - 2a^\top X^\top y + y^\top X(X^\top X)^{-1}X^\top y + y^\top Iy - y^\top X(X^\top X)^{-1}X^\top y \\
&= a^\top (X^\top X)a - 2a^\top X^\top y + y^\top X(X^\top X)^{-1}X^\top y + y^\top (I - X(X^\top X)^{-1}X^\top)y,
\end{aligned} \tag{5}$$

where I is the identity matrix. In (5), we insert $(X^\top X)^{-1}X^\top X$ in appropriate places (since this is just equivalent to the identity matrix I) to obtain

$$\begin{aligned}
z^\top z &= a^\top (X^\top X)a - 2a^\top X^\top y + y^\top X(X^\top X)^{-1}X^\top y + y^\top (I - X(X^\top X)^{-1}X^\top)y \\
&= a^\top (X^\top X)[(X^\top X)^{-1}X^\top X]a - 2a^\top [X^\top X(X^\top X)^{-1}]X^\top y + y^\top X(X^\top X)^{-1}X^\top y \\
&\quad + y^\top (I - X(X^\top X)^{-1}X^\top)y \\
&= \underbrace{(a^\top (X^\top X) - y^\top X)(X^\top X)^{-1}(X^\top Xa - X^\top y)}_{\text{quadratic form}} + y^\top (I - X(X^\top X)^{-1}X^\top)y.
\end{aligned} \tag{6}$$

The matrix $X^\top X$ is square and is positive definite, and so is $(X^\top X)^{-1}$. Thus in (6), the expression

$$(a^\top (X^\top X) - y^\top X)(X^\top X)^{-1}(X^\top Xa - X^\top y) \geq 0$$

and the only way it can be made 0 is when $X^\top Xa - X^\top y = 0$, that is, when $a = (X^\top X)^{-1}X^\top y$. Hence in (6), we have

$$z^\top z \geq y^\top (I - X(X^\top X)^{-1}X^\top)y,$$

with equality if and only if $a = (X^\top X)^{-1}X^\top y$. So this says that the choice of an estimator $\hat{a}$ for a where $a = \hat{a} = (X^\top X)^{-1}X^\top y$ minimizes the sum of squared error $z^\top z$ and is known as the Ordinary Least Squares Estimator for a . In (6), the expression $X(X^\top X)^{-1}X^\top y$ actually gives the vector of fitted values and the expression $y^\top (I - X(X^\top X)^{-1}X^\top)y$ is the residual sum of squares, that is $\sum_{i=1}^n (y_i - \hat{y}_i)^2$. Furthermore, the matrix $(I - X(X^\top X)^{-1}X^\top)$ is idempotent¹, that is $(I - X(X^\top X)^{-1}X^\top)(I - X(X^\top X)^{-1}X^\top) = (I - X(X^\top X)^{-1}X^\top)$.

¹A square matrix M is idempotent if $M^2 = M$.

For the simple linear regression model, (1) becomes

$$y_i = a_0 + a_1 x_i + z_i, \quad (7)$$

for $i = 1, 2, \dots, n$ as in Chapter 2. Writing in vector matrix format, (2) becomes

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}, \quad (8)$$

which can be rewritten as

$$y = Xa + z, \quad (9)$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, $X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$, $a = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$ and $z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$. The least squares estimators for the intercept term a_0 and the gradient term a_1 are then given by

$$\begin{bmatrix} \hat{a}_0 \\ \hat{a}_1 \end{bmatrix} = (X^\top X)^{-1} X^\top y, \quad (10)$$

where $X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$ from (9) and can be compared with the corresponding expressions for $\hat{a}_0$ and $\hat{a}_1$ in Chapter 2.

Now we turn our attention to a general autoregressive model of order p , that is, an $AR(p)$ model. We assume that the model is stationary and that it has constant variance. Assuming that we have n observations $x_1, x_2, \dots, x_n$, the model is

$$x_k = a_1 x_{k-1} + a_2 x_{k-2} + \dots + a_p x_{k-p} + z_k, \quad (11)$$

for $k = p+1, p+2, \dots, n$. Note that in fitting an $AR(p)$ model, we effectively lose the first p observations. Writing in vector matrix notation analogous to (2), we obtain

$$\begin{aligned}
\begin{bmatrix} x_{p+1} \\ x_{p+2} \\ \vdots \\ x_n \end{bmatrix} &= a_1 \begin{bmatrix} x_p \\ x_{p+1} \\ \vdots \\ x_{n-1} \end{bmatrix} + a_2 \begin{bmatrix} x_{p-1} \\ x_p \\ \vdots \\ x_{n-2} \end{bmatrix} + \cdots + a_p \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-p} \end{bmatrix} + \begin{bmatrix} z_{p+1} \\ z_{p+2} \\ \vdots \\ z_n \end{bmatrix} \\
&= \begin{bmatrix} x_p & x_{p-1} & \cdots & x_1 \\ x_{p+1} & x_p & \cdots & x_2 \\ \vdots & \vdots & \cdots & \vdots \\ x_{n-1} & x_{n-2} & \cdots & x_{n-p} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} + \begin{bmatrix} z_{p+1} \\ z_{p+2} \\ \vdots \\ z_n \end{bmatrix}.
\end{aligned} \tag{12}$$

Thus (12) has the form

$$\begin{bmatrix} x_{p+1} \\ x_{p+2} \\ \vdots \\ x_n \end{bmatrix} = Xa + z, \tag{13}$$

where $X = \begin{bmatrix} x_p & x_{p-1} & \cdots & x_1 \\ x_{p+1} & x_p & \cdots & x_2 \\ \vdots & \vdots & \cdots & \vdots \\ x_{n-1} & x_{n-2} & \cdots & x_{n-p} \end{bmatrix}$, $a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix}$ and $z = \begin{bmatrix} z_{p+1} \\ z_{p+2} \\ \vdots \\ z_n \end{bmatrix}$. For short hand notation,

we denote $x = \begin{bmatrix} x_{p+1} \\ x_{p+2} \\ \vdots \\ x_n \end{bmatrix}$ and we now have

$$x = Xa + z, \tag{14}$$

in comparison to (9). As with the multiple linear regression model, the sum of squared errors is $z^\top z = \sum_{i=p+1}^n z_i^2$ and following the same argument set out in the derivation of the least squares estimators for the entries of a , here the least squares estimates of a are given by

$$\hat{a} = (X^\top X)^{-1} X^\top x. \tag{15}$$

The fitted valules are $X(X^\top X)^{-1} X^\top x$ and $x^\top (I - X(X^\top X)^{-1} X^\top)x$ is the residual sum of squares.

Now let us consider the special case of a 1st order autoregressive model, that is, an AR(1) model. This model is

$$x_k = a_1 x_{k-1} + z_k, \tag{16}$$

for $k = 2, 3, \dots, n$. Writing (16) in vector notation, we have

$$\begin{bmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{bmatrix} a_1 + \begin{bmatrix} z_2 \\ z_3 \\ \vdots \\ z_n \end{bmatrix}. \quad (17)$$

The matrix X in the general formula (13) is simply $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{bmatrix}$, thus $X^\top X = \sum_{i=1}^{n-1} x_i^2$ and

the estimate for a_1 is

$$\hat{a}_1 = (X^\top X)^{-1} X^\top x = \frac{\sum_{i=1}^{n-1} x_i x_{i+1}}{\sum_{i=1}^{n-1} x_i^2}. \quad (18)$$